Chong Zhang

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Updated: September 26, 2022 | Zürich, Switzerland

Education

ETH Zürich — Zürich, Switzerland
MSc in Robotics, Systems and Control — *September 2022 - July 2024 (expected)*

Tsinghua University, *Distinguished Graduate* — Beijing, China
B.Eng in Measurement, Control Technology and Instruments; ***GPA: 93/100*** — *August 2018 - June 2022*
Minor in Data Science and Technology; ***GPA: 95/100***

Research Interests

Applied Machine Learning for Robotics

- **Data-efficient reinforcement learning for robust legged locomotion.**
  Representative Work: Accessibility-Based Clustering for Efficient Learning of Locomotion Skills
- **Data generation for data-driven control.**
  Representative Work: Generating a Terrain-Robustness Benchmark for Legged Locomotion

Research Experience

Exploring Exploration in Reinforcement Learning — ETH Zürich, Switzerland
Project at Robotic Systems Lab (RSL) — *September 2022 - present*

- Trying to enhance exploration in reinforcement learning of perceptive quadrupedal locomotion.
- Supervised by Nikita Rudin and David Höller. Lab Leader: Prof. Marco Hutter. Tutor: Prof. Roland Siegwart.

RoboCup Humanoid — Tsinghua University, China
Undergraduate Developer — *July 2021 - March 2022*

- Led the exploration of reinforcement-learning based motions.
- Contributed to some research papers:
  Custom Sine Waves Are Enough for Imitation Learning of Bipedal Gaits with Different Styles.
  Adaptive Mimic: Deep Reinforcement Learning of Parameterized Bipedal Walking from Infeasible References.
- Contributed to the ongoing design of a new humanoid platform.

Efficient Learning of Robot Locomotion Skills — University of Edinburgh, Scotland
Research Internship, advised by Dr. Alex Li — *June 2021 - September 2021*

- Followed up the group's Science Robotics paper: Multi-Expert Learning of Adaptive Legged Locomotion.
- Achieved efficient learning of robot locomotion skills, including robust fall recovery and backflipping.
- Oral presentation in ICRA 2022: Accessibility-Based Clustering for Efficient Learning of Locomotion Skills.

Modelling and Prediction of Ship Motions — Tsinghua University, China
Research Assistant, advised by Dr. Peida Hu — *October 2019 - June 2021*

- Achieved 10-Hz accurate ship motion prediction for future 8 s with online learning.
- Got top-level funding from the National Training Programs of Innovation and Entrepreneurship (hosted by the Ministry of Education in China).
- Technical Report: Modeling of Coupling Effects in Neural Networks for Ship Motion Prediction.

Selected Extracurricular Projects

Generated Terrain-Robustness Benchmark for Legged Locomotion | *GitHub* — Summer 2022

- Combining terrain authoring and active learning to generate a dataset of diverse, challenging, and realistic unstructured terrains in simulation.
- The paper explaining the methodology: https://arxiv.org/abs/2208.07681.

Distributed Motion Planning and Cooperative Coverage for UAV Swarms Spring 2022

- Achieved real-time motion planning of large-scale UAV swarms in parallel via distributed trajectory optimization.
- Solving the NP-hard cooperative coverage problem in real time via multi-TSP modelling and distributed deep reinforcement learning.
- A 60-page Bachelor thesis. Drop me an email if you are interested.

Others: Thinking or Doing what I want to do **Always**

- Built a minimal auto-driving system from scratch. Spring 2021
- Developed a motor driver board with encoders. Spring 2021
- Won the second prize in a national big data challenge final. Summer 2020
- Built a 5-DoF catapult quadrupedal robot weighing < 500 g that can walk and turn. Fall 2019
- Developed a robotic arm system tracking my hand in real time via monocular vision. Summer 2019
- Developed a fingerprint storage box for my stuff. Spring 2019

Service

Reviewer for Academic Journals and Conferences

- 2022 American Control Conference (ACC)
- 2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)
- IEEE Robotics and Automation Letters (RA-L)

Voluntary Experience

Student Association of Science and Technology in Department Tsinghua University, China

Vice President, after 2 years being a regular member (2019-2021) *March 2021 - March 2022*

- The association provides funding, engineering training, and competitions for undergraduates.
- I worked as the leader of the committee for undergraduate competitions in the department.
- I developed my leadership and communication skills.

Skills

Software: Mechanical Design (CAD/Solidworks), Photoshop, NI Labview, NI Multisim, MySQL, ROS

Skill: Coding (Python/Matlab/C++), PCB Design, Embedded System Development, Computer Vision

Languages: Chinese (Native), English (C1), German (Beginner)

Hobbies

Sports: Fencing (Sabre) 🤺, Handball 🤾, Tennis 🎾

And: 🔴Pokémon🔴!

Btw, I LOVE ROBOTS 🤖, FOREVER!!!